Watin Promfiy

DATA SCIENCE / AI ENGINEER / DATA ENGINEER

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PROFESSIONAL SUMMARY

Fourth-year data science undergraduate at Musashino University and KMITL (GPA 3.87), and master's candidate at Musashino University's Graduate School of Data Science, with industry experience in machine learning, LLMs, and predictive maintenance at Asia Energy Tech, Bot and Life, and KBTG, and teaching roles at AIAT, KMITL, and DGA. Author of nine international conference papers accepted in 2026 (eight as first author) and a four-time competition champion, including the BDI Hackathon Royal Cup.

EDUCATION

Master's Program in Data Science (Candidate) — Expected start 2027
Musashino University, Japan · Graduate School of Data Science · Passed graduate entrance examination

Bachelor's Program in Data Science (Year 4) — Sep 2026 – Present
Musashino University, Japan · Faculty of International Data Science

B.Sc. in Information Technology (Data Science and Business Analytics) — 2023 – Present
King Mongkut's Institute of Technology Ladkrabang (KMITL), Thailand · Cumulative GPA: 3.87/4.00

PROFESSIONAL EXPERIENCE

AI Engineer Intern, Asia Energy Tech (AET) — Jul 2026 – Present
- Developed predictive maintenance models for industrial pump sensor data, benchmarking 13 detection methods.
- Built an equipment health dashboard and an LLM maintenance-report generator with automated fact verification.

AI Engineer (Part-Time), Bot and Life — Jul 2025 – Feb 2026
- Engineered a multi-agent platform that automates product banner and marketing asset creation.
- Implemented prompt engineering strategies that expand user inputs in context to improve generated content.

LLM & Prompt Engineer Intern, Bot and Life — Nov 2024 – Feb 2025
- Architected an LLM-powered task management system with memory for long-term conversational context.
- Developed prompt frameworks to minimize hallucination and enforce structured output compliance.

Data Science Intern (Apprentice Program), KBTG (K-LAB) — Jul – Dec 2024
- Developed a PyTorch model estimating roof surface area and solar energy potential from satellite imagery.
- Engineered a condominium similarity engine using geospatial data and deep learning for property valuation.

TEACHING EXPERIENCE

Teaching Assistant, King Mongkut's Institute of Technology Ladkrabang (KMITL) — Jun – Sep 2026
Applied Machine Learning and Data Warehouse courses (DSBA8)
- Developed weekly labs on scikit-learn machine learning and Airflow, dbt, and PostgreSQL data warehouse pipelines.
- Built a student progress dashboard for lab grades and a synthetic retail dataset for a CRISP-DM assignment.

Teaching Assistant, Digital Government Development Agency (DGA) — May – Aug 2026
e-Government Executive Program, Cohorts 1–3
- Mentored participants in analyzing Thai open government data with Python, Plotly, and machine learning models.
- Compiled a public data visualization catalog of the interactive dashboards produced across three cohorts.

Teaching Assistant, Artificial Intelligence Association of Thailand (AIAT) — Apr – Jul 2025; May – Jul 2026
Super AI Engineer Program, Seasons 5 and 6
- Delivered curriculum from machine learning to LLMs, including LLM inference and LoRA/QLoRA sessions on LANTA.
- Mentored participants in Python debugging and led paper-reading sessions on state-of-the-art AI research.

TECHNICAL SKILLS

AI & LLMs: LLM fine-tuning (LoRA/QLoRA), RAG, VLMs, MoE, multi-agent systems, prompt engineering, MLX
Machine Learning: Transformers, CNNs, LSTMs, gradient boosting, anomaly detection, time-series forecasting
Data Science: Statistical modeling, feature engineering, geospatial analysis, knowledge graphs, Plotly
Data Engineering: ETL/ELT, PySpark, Airflow, Kafka, dbt, PostgreSQL, BigQuery, Snowflake
MLOps & Cloud: FastAPI, Docker, inference optimization, vector databases, HPC, AWS, GCP, Huawei Cloud
Programming: Python, SQL, TypeScript, Git, Linux
Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, LlamaIndex, Next.js, n8n

PUBLICATIONS

Publication Summary: 9 accepted international conference papers in 2026 (8 as first author) and 2 journal manuscripts in preparation.

Peer-Reviewed International Conference Papers (Accepted, 2026)

[1] **W. Promfiy**, P. Boonprasart, V. Sornlertlamvanich, and S. Poomrittigul, "HIKARI: A Multi-Stage Vision-Language Prototype with RAG-in-Training for Medical Lesion Diagnosis," in *Proc. International Technical Conference on Circuits/Systems, Computers, and Communications* (ITC-CSCC), 2026.

[2] **W. Promfiy**, T. Jaempinit, S. Koseeyaumporn, and S. Tarnpradab, "Yeeping-Eco: A Knowledge Graph RAG Framework with Predictive PM2.5 Routing," in *Proc. International Conference on Big Data Analytics and Practices* (IBDAP), 2026.

[3] S. Koseeyaumporn, **W. Promfiy**, S. Saiprom, and S. Tarnpradab, "MANJU: A Multi-Agent Framework for Natural Just-in-Time Understanding in Thai," in *Proc. International Joint Conference on Computer Science and Software Engineering* (JCSSE), 2026.

[4] **W. Promfiy** and S. Poomrittigul, "Agents Optional, Calibration Essential: Efficient Medical Question Answering with Open Language Models," in *Proc. Biomedical Engineering International Conference* (BMEiCON), 2026.

[5] **W. Promfiy**, S. Poomrittigul, and W. Buavirat, "Deep-Unfolded Compressed Sensing for Energy-Efficient Wearable ECG: Diagnostic Preservation, Learned Binary Sensing, and Cross-Dataset Generalization," in *Proc. International Conference on Information Technology and Electrical Engineering* (ICITEE), 2026.

[6] **W. Promfiy** and A. Kuiprasert, "ZoonoMoE: A Voice-Driven Zoonotic Disease Surveillance System via Mixture-of-Experts Routing and Domain-Partitioned Retrieval-Augmented Generation," in *Proc. International Conference on Neural Information Processing* (ICONIP), 2026.

[7] **W. Promfiy**, S. Jitraveewon, and P. Thusaranon, "Mahatsachan or Mirage? Null-Matched Auditing of Minimal Affective Prompt Stimuli in Multilingual LLMs," in *Proc. International Joint Symposium on Artificial Intelligence and Natural Language Processing* (iSAI-NLP), 2026.

[8] **W. Promfiy**, C. Sirikayon, P. Tuwanut, and S. Poomrittigul, "PREFER: Persistent-Frequency KV-Cache Eviction for LLM Prefix Caching," in *Proc. International Joint Symposium on Artificial Intelligence and Natural Language Processing* (iSAI-NLP), 2026.

[9] **W. Promfiy**, A. Kuiprasert, and L. Pradittasnee, "Quantization Calibration and Speculative Decoding for On-Device Thai LLMs: A Measurement Study," in *Proc. International Conference on Information Technology* (InCIT), 2026.

Journal Manuscripts in Preparation

[10] **W. Promfiy** (first author), "Calibrate in the State You Deploy: State-Matched Post-Training Quantization for LLMs with Compressed KV Caches," manuscript in preparation.

[11] S. Poomrittigul, D. Rungraungrayabkul, R. Unnaha, W. Sarakhon, **W. Promfiy**, B. Thanathornwong, S. P. Khovidhunkit, and T. Treebupachatsakul, "An Expert System for Oral Potentially Malignant Disorder Screening That Fuses Photographic Image Classification with Natural Language Processing of the Medical History," manuscript in preparation. Contribution: Writing – original draft.

HONORS & AWARDS

Winner (Royal Cup), BDI Hackathon – AI and Data Innovation for Smart Tourism (BDI, CMU) 2025
- Knowledge-graph recommendation engine that generates personalized Chiang Mai itineraries.

Winner (Best Tourism Idea), ReTourism Lab (Tourism Authority of Thailand) 2025
- AI travel companion for Chiang Mai that turns local data into personalized, real-time recommendations.

Winner, Krungsri UniVerse × KMITL Hackathon – Mule Hunt Challenge (Krungsri, KMITL) 2025
- Mule-account network tracing with web crawling, graph visualization, and an LLM query interface.

Winner, ExxonMobil Bootcathon – GenAI (ExxonMobil) 2025
- Agentic n8n workflow with text-to-Pandas natural-language data exploration.

2nd Runner-Up, AI Thailand Hackathon (NECTEC) 2025
- DAMZ: zero-shot object and action detection from Thai and English natural-language commands.

2nd Runner-Up, Huawei ICT Competition – Cloud Track (Huawei Cloud) 2025
- Top-three national placement for cloud-native architecture and AI/LLM system design on Huawei Cloud.

Finalist (Top 10), JUMP THAILAND Hackathon (AIS Academy, MSDHS) 2025
- Inclusive data-annotation and career-path platform that creates data-labeling jobs for older adults.

Silver Medal, Super AI Engineer Season 4 (AIAT) 2024
- Computer vision and predictive modeling solutions, including solar-potential and real estate projects at KBTG.

1st Runner-Up, Krungsri UniVerse × KMITL Hackathon (Krungsri, KMITL) 2024
- AI for Autism: personalized learning content generation for children with autism.

Honorable Mention, AI Thailand Hackathon (NECTEC) 2024
- Thai CV Analyzer for resume evaluation and job matching through reusable APIs.

Finalist (Top 15), JUMP THAILAND Hackathon (AIS Academy, MSDHS) 2024
- Transformer-based image captioning with text-to-speech for visually impaired users.

1st Runner-Up, Innovation for Crime Combating (I4C) 2023
- Deepfake video detection pipeline using a CNN ensemble for digital forensics.

SELECTED PROJECTS

RegMap AI – Regulatory Intelligence Platform UN ESCAP RDTII · 2026
- Mapped ASEAN digital-trade and data-protection laws to RDTII indicators using multilingual OCR and Llama-3.3-70B.

KhonKaen Health × City Intelligence Platform BDI Hackathon · 2026
- Unified clinical, phenomics, and smart-city data in a knowledge graph with a cross-domain AI agent.

Proactive Welfare Link (PWL) Hack for Right · 2026
- Automated welfare-tier assignment from ICD-10 diagnoses and Barthel ADL scores with PDPA-compliant tokenization.

Nemotron-Personas-Thailand NVIDIA × CMKL · 2026
- Generated 300,000 synthetic Thai population records aligned with official statistics using NVIDIA NeMo Data Designer.

JaiVoice – On-Device Thai Voice Assistant 2026
- Built a real-time, interruptible Thai voice assistant running fully on a MacBook (MLX Whisper, Qwen, cloned-voice TTS).

DiscordFlow – ML Experiment Logger Python library · 2026
- Developed a Python library that asynchronously logs training metrics, parameters, and artifacts to Discord.

Agentic AI CV Generator 2026
- Built a web app that turns text into LaTeX CVs with Qwen3-32B, a validation agent, and job matching.

Skin Disease Image Scraper 2026
- Built a scraper that collects up to 8,900 images across 178 skin diseases for medical image classification.

Multi-Modal Deep Learning for Thai Fossil Classification KMITL · 2025
- Developed a multi-modal deep learning pipeline for fossil classification and paleontological analysis in Thailand.

Home Credit – Credit Risk Model Stability KBTG × AIAT Competition
- Built gradient-boosting default-risk models with Gemini-assisted analysis, focused on stability over time.

Table-Based Thai Question Answering Super AI Engineer
- Built question answering over tables using retrieval-augmented generation and LLM fine-tuning.

Other projects: MacBook Vibration Player (Apple Silicon accelerometer), Thai TTS Discord bot, IoT sensor REST API (Hono, PostgreSQL), and OCR-based file organizer (Naive Bayes).

ACADEMIC SERVICE

External Reviewer (Subreviewer): International Conference on Neural Information Processing (ICONIP 2026)